# ROS 2 Model Setup Guide

DR-xcobo 2.0 | v1.0 | ROS 2 Jazzy / RViz2

Build the robot description package, open the model in RViz2, and explore its joints with interactive sliders. The ZIP includes the URDF, all 10 STL meshes, a Python launch file, and an RViz2 configuration.

## Environment

**Ubuntu 24.04 LTS Desktop (Noble) + ROS 2 Jazzy Jalisco**
Tools: RViz2, colcon, ament_cmake, and a Bash terminal.

Start with ROS 2 Jazzy Desktop and its package repository already configured. Use a fresh terminal that does not automatically source ROS 1 or another ROS 2 distribution. These instructions use a separate workspace at **~/ros2_ws**.

Target environment, not yet runtime-tested. Package metadata, launch syntax, model structure, mesh references, and RViz2 configuration are checked during packaging. This bundle provides model visualization; it does not include MoveIt, Gazebo, or hardware controllers.

## 1. Install the required tools

```
source /opt/ros/jazzy/setup.bash
sudo apt update
sudo apt install build-essential unzip python3-colcon-common-extensions \
  ros-jazzy-ament-cmake ros-jazzy-robot-state-publisher \
  ros-jazzy-joint-state-publisher ros-jazzy-joint-state-publisher-gui \
  ros-jazzy-rviz2
```

If /opt/ros/jazzy/setup.bash is missing, install ROS 2 Jazzy first using the official instructions linked on page 2. The commands above add the dependencies for this model package.

## 2. Download and build

Download the ROS 2 URDF ZIP and save it in **~/Downloads/**. Extract the complete package and build:

```
mkdir -p ~/ros2_ws/src
unzip ~/Downloads/dr-xcobo-2.0-ros2-urdf-v1.0.zip -d ~/ros2_ws/src
cd ~/ros2_ws
colcon build --symlink-install --packages-select dr_xcobo_description
source install/setup.bash
```

The package path should be **~/ros2_ws/src/dr_xcobo_description/**, with package.xml directly inside it. Keep the lowercase package name, URDF, and meshes together; extract the ZIP only once. Use a separate ROS 2 workspace rather than the ROS 1 catkin workspace.

# Launch & Explore

DR-xcobo 2.0 | RViz2 visualization

## 3. Start RViz2 and the joint sliders

In a fresh Bash terminal, load ROS 2 and the workspace, then launch:

```
source /opt/ros/jazzy/setup.bash
source ~/ros2_ws/install/setup.bash
ros2 launch dr_xcobo_description display.launch.py
```

Two windows should open: **RViz2** and **Joint State Publisher GUI**. Robot State Publisher supplies the robot description and link transforms. ROS 2 does not require roscore.

To display a static pose without the slider window:

```
ros2 launch dr_xcobo_description display.launch.py gui:=false
```

## 4. Inspect the model

**Camera:** use RViz2's Move Camera tool to orbit, pan, and zoom.
**Joints:** move the sliders in the separate GUI to inspect joint direction and range.
**Frames:** select **Add > TF** in RViz2 to show the link coordinate frames.
**Stop:** press **Ctrl+C** in the launch terminal.

The model has 10 links and 9 joints: 6 revolute, 2 prismatic, and 1 fixed. The finger joints are independent. The URDF geometry and joint properties are preserved; mesh resource paths use the new ROS 2 package name.

## Troubleshooting

**Package not found:** check the package folder, rebuild with colcon, and source ~/ros2_ws/install/setup.bash in the current terminal.

**No robot model:** enable RobotModel; set Description Source to Topic and Description Topic to /robot_description. The topic durability should be Transient Local.

**Frame or mesh errors:** use base_link as the Fixed Frame, keep the full meshes folder, and check the launch terminal for errors.

**Missing sliders:** install ros-jazzy-joint-state-publisher-gui and launch without gui:=false. Start from a graphical Ubuntu desktop session.

Model scope: original movable joints have zero effort and velocity limits. Review these before motion planning, controller use, or physical simulation. This model display package does not command the physical robot.

## Resources

Course resources | ROS 2 URDF ZIP
Official Jazzy installation guide | Official documentation source
Robot State Publisher | RViz2